

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 331: Automata & Computability

Theory Assignment 03 | Spring 2026 | Total Marks : 40

Answer **all** the following questions.

Figures in the right margin indicate total marks.

Deadline: 27th April, Monday, 2026

Q1.	Use the pumping lemma to prove the following languages are not regular: A. $L_1 = \{w \in \{0, 1\}^* : 0^i 1^j, \text{ where } i \geq j\}$ B. $L_2 = \{w \in \{0, 1\}^* : ww^R, \text{ where } R \text{ denotes the reversed string of } w\}$ C. $L_3 = \{w \in \{0, 1\}^* : w \text{ is not a palindrome}\}$ D. $L_4 = \{w \in \{a, b, c\}^* : w \text{ is } a^i b^j c^{k+2}, \text{ where } i = k; i, j, k \geq 0\}$ E. $L_5 = \{w \in \{0, 1\}^* : w \text{ is } 1^m, \text{ where } m \text{ is a prime number}\}$	15
Q2.	Write a CFG for the following languages: A. $L_1 \rightarrow \{w = 0^n 1^m 2^m 3^n \mid n, m > 0\}, \Sigma = \{0, 1, 2, 3\}$ B. $L_2 \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}, \Sigma = \{0, 1, 2\}$ C. $L_3 \rightarrow \{w_1 \# w_2 \mid \text{number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}, \Sigma = \{0, 1\}$ D. $L_4 \rightarrow \{\text{length of } w \text{ is even}\}, \Sigma = \{0, 1, 2\}$ E. $L_5 \rightarrow \{w = 0^i 1^k 2^j \mid i = j; i, j \geq 1; i \text{ is a multiple of two and } k \text{ is a multiple of three}\}, \Sigma = \{0, 1, 2\}$	15
Q3.	Given the following context-free grammar, $\Sigma = \{a, b\}$ $S \rightarrow PE$ $P \rightarrow aPa \mid bPb \mid a \mid b$ $E \rightarrow aaE \mid abE \mid baE \mid bbE \mid \epsilon$ A. Give a leftmost derivation for the string “<i>abababb</i>” and sketch the parse tree corresponding to that derivation. B. Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (A)) for the same string “<i>abababb</i>”.	10